

Hereditary Congenital Facial Paresis Type 3 (HCFP3) — Disease Characteristics Report

MONDO: MONDO:0013880 · **OMIM:** #614744 · **Category:** Mendelian (autosomal recessive)

Evidence base: HCFP3 is an ultra-rare Mendelian disorder described in only a handful of families worldwide. Consequently, much of the mechanistic detail is derived from the mouse model (*Hoxb1*^{-/-}) and from a small number of human case reports/series. Evidence source types are flagged throughout as **[human clinical]**, **[mouse]**, **[in vitro]**, or **[computational]**. Where a canonical resource (OMIM/Orphanet/HPO) would normally supply information but no primary datum was retrievable in this investigation, the item is marked *"not established / not retrieved."*

1. Disease Information

HCFP3 is a **congenital cranial dysinnervation disorder (CCDD)** characterized by non-progressive, usually **bilateral weakness/paralysis of the muscles of facial expression** due to maldevelopment of the **seventh cranial nerve (facial nerve, CN VII)** and its brainstem motor nucleus. It is present from birth and frequently accompanied by hearing loss, strabismus, feeding difficulties, and orofacial/ear dysmorphism. **[human clinical]**

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Key identifiers - **MONDO:** MONDO:0013880 - **OMIM:** 614744 (phenotype) - **Gene:** HOXB1 — OMIM 142968; HGNC:5111; NCBI Gene 3211; Ensembl ENSG00000120094; UniProt P14653; locus 17q21.32 - **Orphanet:** Hereditary congenital facial paresis (ORPHA:91517 group; HCFP3

subtype) - **ICD-11:** LA05 / 8B88.0-type congenital cranial nerve / facial nerve disorders (congenital facial palsy); **ICD-10:** Q07.8 (other specified congenital malformations of nervous system) — mapping approximate. - **MeSH:** "Facial Paralysis"; "Cranial Nerve Diseases"; related MeSH "Möbius Syndrome" (differential).

Synonyms / alternative names: HCFP3; Hereditary congenital facial paresis, type 3; Congenital facial palsy, HOXB1-related; Facial paresis, hereditary congenital, 3; (broader group) hereditary congenital facial palsy.

Data source type: Aggregated disease-level knowledge (OMIM/Orphanet/HPO) plus individual patient case reports (EHR-style descriptions of small families), **not** large registry/EHR cohorts.

2. Etiology

Primary cause (genetic): Biallelic (homozygous or compound heterozygous) pathogenic variants in **HOXB1**, a homeodomain transcription factor. HOXB1 is "the only known causative gene for HCFP" (HCFP3). **[human clinical]**

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Genetic risk factors - Causal variants: HOXB1 loss-of-function and homeodomain missense alleles (Section 4). - **Consanguinity:** Strong contributor — recessive disease enriched in consanguineous/endogamous families (e.g., Moroccan consanguineous family; German-American conservative isolate). **[human clinical]**

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- **Founder effect:** A founder Arg207Cys allele segregates in a "conservative German American population." **[human clinical]**

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- **Susceptibility/modifier loci:** Not established.

Environmental risk factors: None identified. HCFP3 is a monogenic developmental disorder; unlike acquired/syndromic congenital facial palsy, it is not attributable to birth trauma, teratogens (e.g., misoprostol/Möbius association), or ischemia. Congenital onset means the causal event is embryonic hindbrain patterning.

Protective factors (genetic/environmental): None established. In a recessive disorder, a single wild-type HOXB1 allele is effectively protective (carriers are unaffected —).

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Gene–environment interactions: None documented; disease is fully genetically determined by biallelic HOXB1 dysfunction.

3. Phenotypes

Phenotype	Type	Onset	Severity/ Course	Frequency	HPO suggestion
Bilateral facial (CN VII) palsy	Clinical sign	Congenital	Non-progressive, stable; variable severity	Defining (~100%)	HP:0010628 Facial palsy; HP:0000260 (bilateral)
Impaired facial expression / weak eye closure, drooling	Physical manifestation	Congenital	Stable	High	HP:0000317 Facial features / HP:0000508 Ptosis (variable)
Feeding/sucking difficulties (infancy)	Symptom	Neonatal	Often improves	Common	HP:0011968 Feeding difficulties; HP:0002033 Poor suck
Hearing loss (sensorineural and/or conductive)	Lab/clinical sign	Congenital	Stable	Frequent	HP:0000365 Hearing impairment
Strabismus	Clinical sign	Congenital	Stable	Frequent	HP:0000486 Strabismus
Ear malformations / low-set or dysmorphic ears	Physical	Congenital	Stable	Variable	HP:0000377 Abnormal pinna morphology; HP:0000369 Low-set ears
Orofacial dysmorphism (e.g., upturned nose, upper-lip/	Physical	Congenital	Stable	Variable	HP:0000463 Anteverted nares

Phenotype	Type	Onset	Severity/ Course	Frequency	HPO suggestion
philtrum changes)					
Facial nerve axonal neuropathy (± nerve hypoplasia)	Lab (electrophysiology/ imaging)	Congenital	Stable	Reported	HP:0009830 Peripheral neuropathy
Preserved eye abduction (CN VI intact) — distinguishes from Möbius	Discriminating sign	—	—	Characteristic	(absence of HP:0031747- type abducens palsy)

[human clinical] sources:

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Note Brugnoli 2025

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describes a case with **facial nerve axonal neuropathy without nerve hypoplasia** and **preserved ocular motor skills**, broadening the imaging/electrophysiology spectrum.

Quality-of-life impact: Facial diplegia impairs emotional expression, eye protection (risk of exposure keratopathy), articulation, oral competence (drooling, feeding), and social/psychological well-being; hearing loss adds communication/developmental burden. **Measured evidence:** subjects with congenital facial weakness (includes HCFP) have significantly worse oral health-related quality of life than matched controls — **OHIP-14 13.11 ± 8.11 vs 4.46 ± 4.98** (Liberton 2024,

[P 38791829](#)).

No HCFP3-specific EQ-5D/SF-36 data exist (broader CFW/facial-palsy inference).

4. Genetic / Molecular Information

Causal gene: HOXB1 (HGNC:5111; OMIM 142968; 17q21.32) — homeobox transcription factor, HOX family, master regulator of hindbrain antero-posterior patterning. **[human clinical]**

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Pathogenic variants reported | Variant (cDNA / protein) | Type | Zygosity / origin | Functional consequence | Reference | |---|---|---|---|---| | c.619C>T, p.(Arg207Cys) | Missense (homeodomain Arg5) | Homozygous, **founder** (German-American isolate) | Disrupts DNA minor-groove contact; destabilizes HOXB1:PBX1:DNA complex; altered transcriptional activity **[computational + in vitro]** |

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| | p.Arg207His (historical) | Missense (same residue) | Homozygous | Same residue class; altered cofactor/DNA binding | (reviewed in 27144914) | | c.66C>G, p.(Tyr22) | *Nonsense (truncating)* | *Homozygous, consanguineous Moroccan family* | *Loss of function** (first bona fide LOF allele) |

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| | Two novel compound heterozygous variants | (per report) | Compound het, by exome sequencing | Consistent with LOF; axonal neuropathy phenotype |

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- **ACMG classification:** Reported alleles are pathogenic/likely pathogenic; segregation with disease, rarity, and functional data support classification (query ClinVar for current status).
- **Allele frequency:** Causal alleles are extremely rare/absent in gnomAD (private/founder); precise frequencies not retrieved here.
- **Somatic vs germline: Germline** (constitutional developmental disorder).
- **Functional class: Loss of function** unifies missense (impaired DNA/cofactor binding) and nonsense (truncation) alleles — "all HOXB1 variants reported so far also have severe impact on activity of this transcriptional regulator." **[human clinical / in vitro]**

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Modifier genes: None established. Candidate interacting partners at protein level include **PBX1** (obligate HOX cofactor) and **MEIS** proteins; HOXA1 is a paralog acting in overlapping hindbrain programs (HOXA1 mutations cause a distinct CCDD, Bosley-Salih-Alorainy/Athabaskan brainstem dysgenesis).

Epigenetic information: No disease-specific methylation/histone data for HCFP3 (not established).

Chromosomal abnormalities: None characteristic for HCFP3 (single-gene disorder; no recurrent CNV/translocation).

HCFP locus landscape (nosological context). HCFP is genetically heterogeneous: - **HCFP3 (OMIM 614744):** *HOXB1*, 17q21.32, **autosomal recessive** (this disease). - **HCFP1 (OMIM 601471):** maps to 3q21-q22; resolved in 2023 as **heterozygous duplications of a neuron-specific GATA2 regulatory region (two enhancers + one silencer) and noncoding silencer SNVs** (some impair NR2F1 binding) — autosomal dominant. A humanized mouse extends *Gata2*, favoring inner-ear efferent over facial-branchiomotor fate, rescued by conditional *Gata3* loss. **[human/mouse]**

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- **HCFP2 (OMIM 604185):** maps to 10q21.3-q22.1; gene not yet identified — autosomal dominant.

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- **Other dominant HCFP gene:** *MEPE* frameshift p.(Gln425Lysfs38) with mixed hearing loss in a four-generation family. **[human clinical]***

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Convergent mechanism across subtypes: all HCFP genes act on **maldevelopment of rhombomere-4-derived facial branchiomotor neurons (FBMNs)** — the same lineage disrupted by *HOXB1* loss — so a CFW/HCFP diagnostic gene panel should include *HOXB1*, the *GATA2* regulatory region, and *MEPE*.

5. Environmental Information

- **Environmental factors:** None causal (purely genetic developmental disorder).
- **Lifestyle factors:** Not applicable to disease causation.

- **Infectious agents:** None. (Distinguish from acquired facial palsy such as Bell's palsy [HSV], Lyme disease, otitis media — these are **differential diagnoses**, not HCFP3.)

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic HOXB1 loss-of-function** (nonsense truncation, or homeodomain missense that cripples DNA/cofactor binding) **results in** absent or non-functional HOXB1 transcription factor in the embryonic hindbrain. **[human clinical/in vitro]**

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2. This **leads to** failure to **maintain rhombomere 4 (r4) identity** — r4 patterning is initiated but not sustained (molecular markers appear then fade). **[mouse]**

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3. Loss of r4 identity **results in** mis-specification of r4-derived **facial branchiomotor (FBM) neurons** and **contralateral vestibuloacoustic (CVA) efferent neurons**. **[mouse]**

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4. Mis-specified motor neurons **differentiate but fail to migrate** to their normal positions; instead they form an atypically migrating motor nucleus. **[mouse]**

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5. This aberrant development **leads to subsequent loss of the facial motor nerve** (hypoplasia/aplasia of CN VII nucleus and nerve). **[mouse]**

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6. Absent/deficient facial motor innervation **results in** congenital **bilateral facial muscle paralysis** (impaired expression, eye closure, oral competence). **[human clinical]**

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7. **Branch (auditory):** disruption of r4-derived vestibuloacoustic efferents and inner-ear developmental programs **contributes to hearing loss**. **[mouse/human]**

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(inferred link between CVA lineage and auditory deficit)

8. **Branch (ocular):** associated **strabismus** arises from developmental co-involvement, though eye **abduction (CN VI)** is characteristically **spared**, distinguishing HCFP3 from Möbius syndrome. **[human clinical]**

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9. **Branch (variant-specific):** some human cases show **facial nerve axonal neuropathy without frank nuclear/nerve hypoplasia**, indicating the lesion can manifest as an axonal/dysinnervation phenotype rather than complete agenesis. **[human clinical]**

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The human founder mutation "recapitulates the phenotype of Hoxb1^{-/-} mice," directly bridging the mouse mechanism to human disease. **[human/mouse]**

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Upstream regulatory context (why r4, and what maintains HOXB1)

HOXB1's r4-restricted expression is set up by a **retinoic acid (RA; CHEBI:15367) morphogen gradient**: RA produced by **Raldh2** in paraxial mesoderm induces Hoxb1 up to r4 through 3'/5' RA-response elements, while **Cyp26** RA-degrading enzymes sharpen the r3/r5 boundaries. In *Raldh2*^{-/-} embryos, Hoxb1-expressing cells scatter instead of forming a defined r4. **[mouse]**

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Once induced, HOXB1 **maintains its own expression** via an r4 **autoregulatory element (b1-ARE)** driven by **HOXB1:PBX1** heterodimers, further tuned by **TALE cofactors (PREP1/MEIS)** and SOX/OCT complexes; HOXB1 is a stronger activator than its paralog HOXA1. **[in vitro/mouse]**

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Pathogenic homeodomain variants (Arg207Cys) cripple precisely the PBX1/DNA interactions that sustain this loop — mechanistically linking genotype to the "identity-not-maintained" phenotype in step 2 above.

Category checklist mapped to the chain

- **Molecular pathways:** Upstream **retinoic-acid signaling** (Raldh2/Cyp26 gradient) patterns r4; the core lesion is in the **HOX/PBX–MEIS/PREP transcriptional regulatory network** governing hindbrain segmentation and cranial motor neuron identity (Reactome "Activation of HOX genes"; developmental gene-regulatory network). No classic signaling cascade (Wnt/MAPK/mTOR) is the proximal lesion; the defect is transcription-factor–level. **[mouse/in vitro]**

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- **Protein dysfunction:** Homeodomain Arg207 (= conserved Arg5) normally contacts thymine in the DNA minor groove via hydrogen bonding/electrostatics; mutation destabilizes the **HOXB1:PBX1:DNA** ternary complex → altered target-gene transcription. Nonsense allele → no functional protein. **[computational + in vitro]**

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- **Cellular processes:** Cell-fate specification, **neuronal migration** (tangential migration of branchiomotor neurons), maintenance of segmental identity, neuronal survival. **[mouse]**

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- **Immune / metabolic / oxidative mechanisms:** Not involved (developmental, non-inflammatory, non-metabolic).
- **Molecular profiling / single-cell / omics:** No disease-specific transcriptomic/proteomic/metabolomic datasets for HCFP3 (not established); mechanistic data come from marker/lineage studies in mouse.

Upstream vs downstream: HOXB1 loss (upstream) → r4 identity failure → FBM/CVA mis-specification/migration failure → CN VII nucleus/nerve loss → facial paralysis (downstream clinical readout).

Ontology suggestions: GO:0021610 facial nerve morphogenesis; GO:0021612 facial nerve structural organization; GO:0001764 neuron migration; GO:0048704 embryonic skeletal segment specification; GO:0006357/0006355 regulation of transcription by RNA Pol II; GO:0021546 rhombomere development. **CL:** CL:0000100 motor neuron; CL:0011001 spinal/branchiomotor motor neuron. **UBERON:** UBERON:0005396 rhombomere 4; UBERON:0001647 facial nerve; UBERON:0002894 hindbrain/rhombomere.

7. Anatomical Structures Affected

- **Organ/system level:** Central & peripheral **nervous system** — brainstem (pons/medulla, hindbrain) and cranial nerve VII; **musculoskeletal** (muscles of facial expression, secondary); **auditory system** (inner ear/CN VIII pathway); **visual/oculomotor** (strabismus).
- **Primary structure:** Facial motor nucleus and **facial nerve (CN VII)** — UBERON:0001647.
- **Secondary involvement:** Muscles of facial expression (denervation), ear structures (hearing loss, pinna dysmorphism), extraocular alignment.
- **Tissue/cell level:** **Neural tissue**; specifically **branchiomotor motor neurons** of rhombomere 4 (CL:0000100 motor neuron), vestibuloacoustic efferent neurons.
- **Subcellular level:** **Nucleus** (GO:0005634) — site of HOXB1 transcription-factor action on chromatin/DNA.
- **Localization & lateralization:** Bilateral (often symmetric) facial involvement; brainstem hindbrain r4 territory. **[human clinical]**

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8. Temporal Development

- **Onset: Congenital** (embryonic hindbrain maldevelopment); facial weakness evident at birth/neonatal period (feeding/sucking difficulty, incomplete eye closure). **[human clinical]**
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- **Onset pattern:** Static/developmental (not acute or acquired).
- **Progression: Non-progressive / stable** over life (a dysinnervation/malformation, not a degenerative process). No disease staging.
- **Course:** Lifelong, chronic, stable. Feeding difficulties may improve with maturation; facial weakness persists.
- **Remission:** None spontaneous; deficits are structural. Interventions are supportive/reconstructive (Section 12).
- **Critical period:** The vulnerable/opportunity window is **embryonic hindbrain segmentation** (r4 identity maintenance); postnatally the developmental lesion is fixed.

9. Inheritance and Population

- **Inheritance pattern: Autosomal recessive** (HCFP3). Contrast: HCFP1 (OMIM 601471) and HCFP2 (OMIM 604185) are autosomal dominant. **[human clinical]**
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- **Penetrance:** Appears complete in reported biallelic individuals; heterozygous carriers are unaffected. **[human clinical]**
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- **Expressivity:** Variable (severity of hearing loss, strabismus, dysmorphism, presence/absence of nerve hypoplasia vary between and within families). **[human clinical]**
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- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism:** Not reported.

- **Founder effect:** Yes — Arg207Cys founder allele in a conservative German-American population. **[human clinical]**

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- **Consanguinity:** Prominent contributor (homozygous alleles in consanguineous/endogamous families). **[human clinical]**

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- **Carrier frequency:** Not established (ultra-rare; population-specific).
- **Epidemiology:** HCFP overall is very rare; HCFP3 specifically is reported in only a small number of families/cases worldwide. Precise prevalence/incidence per 100,000 **not established**. Broader hereditary congenital facial palsy is estimated at roughly ~2 per million births (order-of-magnitude, group-level; not HCFP3-specific).
- **Sex ratio:** No strong sex bias expected for an autosomal recessive disorder; formal data not established.
- **Geographic distribution:** Case clusters reported in a German-American isolate (founder), a consanguineous Moroccan family, Russian patients (Murtazina 2023,

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and Italian (Brugnoli 2025). Distribution reflects reporting/founder/consanguinity rather than true endemicity.

10. Diagnostics

Clinical evaluation - Recognition of **congenital, non-progressive, bilateral facial weakness** with preserved eye abduction; assess feeding, eye closure, hearing, ocular alignment, ear/facial morphology. **[human clinical]**

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Genetic testing (definitive) - **Exome sequencing (ES/WES)** is the demonstrated diagnostic modality (used to identify HOXB1 variants). **[human clinical]**

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- **Targeted HOXB1 single-gene / CCDD gene-panel testing** appropriate once phenotype suggests HCFP3. - **Genome sequencing (WGS)** where ES is uninformative. - In individuals with congenital facial paralysis, preserved ocular motor skills, and confirmed facial-nerve axonal neuropathy, "**HOXB1 variants and therefore a diagnosis of HCFP3 should be primarily considered.**" [human clinical]

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Electrophysiology / functional tests - Facial nerve conduction, blink reflex, ± needle EMG: HCFP (and Möbius) subjects show **low-amplitude CN VII responses without other neuropathic or myopathic findings**, distinguishing them from generalized-neuropathy CFW (e.g., TUBB3 polyneuropathy) or myopathic CFW (Carey-Fineman-Ziter). May also show axonal facial neuropathy. [human clinical]

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- **Audiometry / BAER (ABR):** to detect and characterize hearing loss (sensorineural/conductive). - **Ophthalmologic exam:** strabismus, tear-film/exposure assessment.

Imaging - High-resolution MRI (brainstem/CN VII, IAC): grades facial-nerve maldevelopment 0–4 (0 normal → 1 unilateral hypoplasia → 2 unilateral aplasia → 3 bilateral aplasia/hypoplasia → 4 with additional cranial-nerve involvement) and can reveal inner/middle/external-ear anomalies; in HCFP3 the nerve may be **hypoplastic/aplastic OR normal-appearing despite axonal neuropathy** (variable). [human clinical]

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Biomarkers / lab chemistry / biopsy: No specific blood/urine biomarker; diagnosis is clinical + molecular. No pathognomonic histopathology.

Clinical criteria & differential diagnosis: Distinguish from **Möbius syndrome** (adds CN VI abduction deficit), acquired facial palsy (birth trauma, Bell's palsy, infection), **HOXA1-related CCDDs**, **TUBB3-related CFW** (adds generalized sensorimotor axonal polyneuropathy on EDx), **Carey-Fineman-Ziter** and other myopathic CFW (myopathic EDx findings), **CHARGE**

syndrome, hemifacial microsomia/oculo-auriculo-vertebral spectrum, and other syndromic congenital facial weakness. Preserved abduction + **isolated low-amplitude CN VII** pattern without neuropathic/myopathic features favors HCFP. **[human clinical]**

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Screening: Carrier/cascade testing within affected families; prenatal/preimplantation testing feasible once the familial HOXB1 genotype is known. Not part of population newborn screening.

11. Outcome / Prognosis

- **Survival/mortality: Normal life expectancy;** HCFP3 is not life-limiting (no cardiorespiratory/visceral fatal component). Disease-specific mortality negligible.
- **Morbidity/disability:** Chronic functional impairments — impaired facial expression and eye closure (exposure keratopathy risk), speech/articulation difficulty, oral competence/feeding issues in infancy, hearing impairment, strabismus/amblyopia risk. Psychosocial impact from facial diplegia; measurably reduced oral health-related QoL (OHIP-14 13.11 vs 4.46,

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- **Recovery potential:** The primary facial-nerve deficit is structural and **does not resolve spontaneously**; supportive/rehabilitative and reconstructive measures improve function. Feeding difficulties often improve with age.
- **Prognostic factors:** Severity of nerve hypoplasia, degree of hearing loss, presence of associated anomalies; earlier multidisciplinary intervention improves functional/QoL outcomes.

- **Prognostic biomarkers:** None established.

12. Treatment

There is **no disease-modifying or curative therapy**; management is **supportive, rehabilitative, and reconstructive**, delivered by a multidisciplinary team. (No pharmacotherapy targets the underlying transcription-factor defect.)

- **Pharmacotherapy:** No specific drug. Ocular surface protection: **lubricating eye drops/ointments** (artificial tears) to prevent exposure keratopathy. Pharmacogenomics: not applicable.

- **Surgical / interventional:**

- **Facial reanimation surgery — free functional muscle transfer**, most commonly **segmental free gracilis muscle transfer (FGMT)** neurotized by the **masseteric (trigeminal) nerve** or via **cross-face nerve grafts**, is the standard smile-reanimation technique for non-resolving congenital bilateral facial paralysis (the HCFP/Möbius category). **[human clinical]**

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 Pediatric outcomes: ~84% achieve active gracilis contraction with mean **commissure excursion gains ~9.7 mm at 1 year**

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 gains are **maintained/improved at 5–13 years**

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 1-stage and 2-stage bilateral approaches give comparable outcomes

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(Series are Möbius-dominated but the reconstructive principles apply directly to HCFP3.)

NCIT: facial reanimation; free muscle flap transfer; cross-face nerve grafting.

- **Ophthalmic:** strabismus surgery; eyelid procedures (gold-weight implant, tarsorrhaphy) for lagophthalmos/eye protection.

- **Otologic/audiologic:** hearing aids or bone-conduction devices; cochlear implantation if indicated for severe SNHL.
- **Supportive / rehabilitative:** Feeding support in infancy; **speech and language therapy** (articulation, oral competence); **physical/occupational/facial neuromuscular retraining therapy**; early developmental support; ophthalmology and audiology follow-up.
- **Advanced therapeutics (gene/cell/RNA/targeted/immuno):** None developed; congenital developmental fixation of the lesion limits gene-therapy applicability.
- **Experimental treatments / trials:** No HCFP3-specific interventional trials identified (ClinicalTrials.gov — not retrieved for this specific ultra-rare entity).
- **Treatment strategy:** Individualized, phenotype-directed multidisciplinary care (facial reanimation timing, eye protection, hearing habilitation, speech therapy) + genetic counseling.

NCIT suggestions: facial reanimation / nerve graft procedures; strabismus surgery; hearing aid; cochlear implant; supportive/palliative care; physical therapy; speech therapy.

13. Prevention

- **Primary prevention:** Not possible for an inherited developmental disorder other than via **reproductive genetic options** — **genetic counseling**, **carrier testing** in at-risk families/ consanguineous couples, **prenatal diagnosis** and **preimplantation genetic testing (PGT-M)** once the familial HOXB1 variants are known.
- **Secondary prevention (early detection/complication avoidance):** Early recognition to institute **eye protection** (prevent exposure keratopathy), **hearing habilitation** (support language development), and **strabismus/amblyopia management** in the critical visual-development window.
- **Tertiary prevention:** Ongoing corneal protection, feeding/nutrition support, speech therapy, and timely facial reanimation to limit functional and psychosocial complications.
- **Immunization / public-health / environmental / prophylaxis:** Not applicable (non-infectious, non-environmental).
- **Counseling:** Autosomal recessive recurrence risk = 25% for a couple who are both carriers; emphasize consanguinity risk and founder-population screening. **[human clinical]**

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14. Other Species / Natural Disease

- **Taxonomy:** *Homo sapiens* (NCBI Taxon 9606); disease-model *Mus musculus* (Taxon 10090).
- **Orthologous gene:** Mouse **Hoxb1** (NCBI Gene 15407; MGI); zebrafish **hoxb1a/hoxb1b** (ZFIN) with cofactor **pbx4** (*lazarus*). Highly conserved homeodomain (human Arg207 = conserved homeodomain Arg5). **[mouse/zebrafish]**

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- **Natural disease in other species / breeds:** No naturally occurring HOXB1-facial-paresis disease catalogued (OMIA — not retrieved); no VBO breed association.
- **Comparative biology:** The mouse **Hoxb1**^{-/-} phenotype (loss of r4 identity, FBM/CVA mis-migration, facial motor nerve loss) closely mirrors the human disorder — human founder mutation "recapitulates the phenotype of Hoxb1^{-/-} mice," demonstrating deep evolutionary conservation of the hindbrain segmentation/branchiomotor program. **[human/mouse]**

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- **Zoonotic potential / cross-species transmission:** Not applicable (genetic disorder).

15. Model Organisms

- **Primary model:** Mouse — **Hoxb1 knockout (Hoxb1^{-/-})** [mammalian, genetic knockout]. Key resource: MGI. **[mouse]**

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- **Model characteristics / phenotype recapitulation:** Faithfully reproduces the core human mechanism — r4 identity is initiated but not maintained; FBM and CVA neurons are mis-specified and **fail to migrate**, forming an atypically migrating nucleus with **loss of the facial motor nerve** → facial paralysis analog. Explicitly stated to correlate "extensively"

with the human phenotype. **[mouse]**

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- **Second vertebrate model — zebrafish (*Danio rerio*, NCBI Taxon 7955):** *hoxb1a* interacts genetically with *lazarus/pbx4* to control **facial (CN VII) motor neuron migration**; genetic mosaic analysis shows both act **primarily cell-autonomously within the facial motor neurons** (with a minor non-cell-autonomous component). This independently validates the HOX/PBX-driven FBMN migration mechanism disrupted in HCFP3. **[zebrafish]**

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Orthologs: *hoxb1a*, *hoxb1b*, *pbx4*. Resource: ZFIN.

- **Model types available:** Constitutive knockout (Studer 1996); zebrafish morphant/mutant (Cooper 2003); paralog/compound studies (*Hoxa1/Hoxb1*) and knock-in swap experiments exist in the developmental-biology literature (not detailed here). A **humanized HCFP1 (GATA2-regulatory) mouse** exists for the related subtype

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Conditional/humanized HOXB1 alleles — resources via MGI/IMPC (not enumerated in this investigation).

- **Limitations:** Mouse constitutive knockout captures nerve/nucleus maldevelopment but does not fully model human associated features (hearing loss severity, strabismus, human-specific dysmorphism) or the milder axonal-neuropathy-without-hypoplasia human presentations.
- **Applications:** Dissecting hindbrain segmentation, branchiomotor neuron specification/migration, and HOX/PBX transcriptional control; platform for genotype-phenotype correlation.
- **In vitro / computational:** In vitro DNA–protein binding assays and molecular modeling of the Arg207Cys homeodomain–DNA interaction. **[in vitro/computational]**

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- **Resources:** MGI (*Hoxb1*), IMPC/IMSR for alleles.

Summary Answer

Hereditary Congenital Facial Paresis type 3 (HCFP3; OMIM #614744, MONDO:0013880) is an ultra-rare autosomal-recessive congenital cranial dysinnervation disorder caused by biallelic loss-of-function of the hindbrain transcription factor **HOXB1** (17q21.32; e.g., founder p.Arg207Cys and nonsense p.Tyr22). *Loss of HOXB1 prevents maintenance of rhombomere-4 identity, causing mis-specification and failed migration of facial branchiomotor neurons and consequent hypoplasia/dysfunction of the facial motor nucleus and nerve — producing congenital, non-progressive bilateral facial paralysis** with variable hearing loss, strabismus, feeding difficulty and orofacial/ear anomalies, while eye abduction (CN VI) is characteristically spared (distinguishing it from Möbius syndrome). Diagnosis is by exome/HOXB1 sequencing; there is no cure, and management is multidisciplinary and supportive (eye protection, hearing habilitation, speech therapy, facial reanimation) with autosomal-recessive genetic counseling (25% recurrence risk).

Key Findings (with evidence)

1. **Genetic cause:** biallelic HOXB1 variants → HCFP3 (autosomal recessive), the only established HCFP gene

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P 27144914
P 22770981
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2. **Variant spectrum:** founder missense p.Arg207Cys, nonsense p.Tyr22* (LOF), and compound-het alleles; unified by loss of function

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3. **Mechanism:** HOXB1 loss → failed r4 identity maintenance → FBM/CVA neuron mis-migration → facial motor nerve loss → bilateral facial palsy (mouse *Hoxb1*^{-/-};

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4. **Phenotype:** bilateral facial palsy + hearing loss + strabismus + orofacial/feeding anomalies, with preserved eye abduction

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Supported vs. Refuted Hypotheses

- **Supported:** HOXB1 LOF is causal and recessive; mechanism is developmental (r4/branchiomotor); mouse model recapitulates human disease.
- **Refuted / not applicable:** environmental/infectious/metabolic/immune etiology; somatic origin; progressive or degenerative course; gene–environment interaction.

Limitations & Future Directions

- Ultra-rare disease: prevalence/incidence, penetrance/expressivity ranges, sex ratio, and gnomAD carrier frequencies are **not precisely established**.
- Mechanistic branch linking HOXB1/r4 to human hearing loss and strabismus is **partly inferred** from mouse/lineage data.
- No omics (transcriptomic/proteomic) profiling exists for HCFP3 tissue.
- Future work: broaden genotype–phenotype catalog (ClinVar/GeneMatcher), define natural history/QoL, refine imaging/electrophysiology correlates (hypoplasia vs axonal neuropathy), and explore conditional/humanized models.

Evidence classes: [human clinical] case reports/families & surgical series; [mouse] Hoxb1/Raldh2 knockouts; [in vitro] DNA-binding/enhancer assays; [computational] molecular modeling. Primary PMIDs: 22770981 (Webb 2012, founder HOXB1), 27144914 (Vogel 2016, LOF nonsense),

8967950 (Studer 1996, *Hoxb1*^{-/-} mouse), 39235314 (Brugnoli 2025, compound-het), 38203298 (Murtazina 2023, phenotype expansion), 15872003 (Sirbu 2005, *RA*→*Hoxb1* r4), 10654602 (Niederreither 2000, *Raldh2* hindbrain), 11278854 (Di Rocco 2001, b1-ARE *HOXB1:PBX1*), 10654609 (Ferretti 2000, *PREP1/MEIS*), 30556292 (Bell 2019, Möbius differential), 37386251 (Tenney 2023, *HCFP1/GATA2* regulatory), 30287925 (Schrauwen 2019, *MEPE HCFP*), 33389762 (Lehky 2021, electrodiagnostics), 30074067 (Mohammad 2018, MRI grading), 38791829 (Liberton 2024, *OHRQoL*), 12645925 (Cooper 2003, zebrafish *hoxb1a/pbx4*), 30166122 / 33637466 / 40738135 / 33191114 / 40100160 (facial reanimation).

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